

Autocallables_research

June 20, 2026

1 Autocallables: Computing Expected Life

1.1 Quick Refresher on markov Modeling

Concept	Definition	Key criterion / formula	In the autocallable chain
State space	The set of all possible values X_n can take	—	$\{A_1, A_2, A_3, A_4, C_1, C_2, C_3, M_H, M_M, M_L\}$
Transition matrix P	Row-stochastic: $P_{ij} = P(X_{n+1} = j \mid X_n = i)$	every entry ≥ 0 , $\sum_j P_{ij} = 1$	each row of $[Q \ R]$ sums to 1 by construction
Transient state	Visited only finitely many times a.s.; probability of return < 1	$\sum_n P_{ii}^{(n)} < \infty$	A_1, A_2, A_3, A_4
Absorbing state	Special case of recurrent: once entered, never left	$P_{ii} = 1$	$C_1, C_2, C_3, M_H, M_M, M_L$
Recurrent state	Probability of eventual return = 1	$\sum_n P_{ii}^{(n)} = \infty$	none of the <i>non-absorbing</i> states are recurrent — that's what makes the chain absorbing
Communicating class	Maximal set of states all mutually reachable	$i \leftrightarrow j$	each A_n and each absorbing state is its own singleton class — no two-way communication
Irreducibility	Single communicating class; every state reachable from every other	—	fails by design — once absorbed, you can't reach a transient state again
Periodicity	gcd of return times to a state	$d(i) = \gcd\{n : P_{ii}^{(n)} > 0\}$	not meaningful here — transient states are never revisited, absorbing states are period 1 (self-loop)

Concept	Definition	Key criterion / formula	In the autocallable chain
Ergodicity	Irreducible + aperiodic + positive recurrent $\Rightarrow$ unique stationary π , time-avg = ensemble-avg	requires irreducibility	does not apply — absorbing chains are never ergodic; long-run behavior is “stuck at an absorbing state,” not a stationary mix
Stationary distribution π	$\pi P = \pi$, $\sum_i \pi_i = 1$	unique only if irreducible & positive recurrent	here, <i>any</i> point mass on an absorbing state is stationary — degenerate, not informative
Canonical form (absorbing chains)	Reorder transient states first, absorbing last	$P = \begin{pmatrix} Q & R \\ 0 & I \end{pmatrix}$	exactly the Q, R split used in the autocallable formulation
Fundamental matrix N	Expected visits to transient state j starting from i	$N = (I - Q)^{-1}$, well-defined iff $Q^n \rightarrow 0$	$N = I + Q + Q^2 + Q^3$ since Q is nilpotent (chain only moves forward)
Absorption probabilities	$P(\text{absorbed in } j \mid \text{start } i)$	$B = NR$	gives $P(C_n) = \pi_n p_n$, $P(M_H), P(M_M), P(M_L)$
Expected time to absorption	Expected number of steps before hitting an absorbing state	$t = N\mathbf{1}$	expected life calculation
Chapman–Kolmogorov	Multi-step transitions compose	$P^{(m+n)} = P^{(m)}P^{(n)}$	used implicitly when propagating S_t ’s distribution from one observation date to the next
Perron–Frobenius / spectral gap	For irreducible aperiodic chains, second-largest eigenvalue magnitude sets the mixing rate to π	$\lambda_1 = 1 > \lambda_2 $	not relevant for a finite-horizon absorbing chain — there’s no mixing to a stationary distribution to speak of
π	steady-state distribution	satisfies $\pi = \pi P$	a probability vector where each entry corresponds to the long-term likelihood of the process being in a specific state, regardless of the starting conditions.

Reference: ORF309 lecture notes, Claude

Parameters: maturity 8 months, non-call period 2 months, observations every 2 months, autocall barrier 100%, coupon barrier 70%, maturity barrier 60%.

1.2 Setup

Treat the underlying's level relative to its initial value as the random driver. Since it is observed at discrete dates and (under standard GBM-type modeling) each date's level depends only on the *previous* date's level, the process is Markov by construction.

Observation dates: $n = 1, 2, 3, 4$ at $t_n = 2, 4, 6, 8$ months. $n = 1, 2, 3$ are call dates; $n = 4$ is maturity.

Transient states (not yet absorbed): A_1, A_2, A_3, A_4 = "alive, entering observation n ."

Absorbing states: C_1, C_2, C_3 (called at observation n), and M_H, M_M, M_L (matured: full payout / par-only / capital loss).

1.3 State Diagram

1.4 Canonical form and the fundamental matrix

Order transient states $[A_1, A_2, A_3, A_4]$. Since the chain only moves forward:

$$Q = \begin{pmatrix} 0 & 1-p_1 & 0 & 0 \\ 0 & 0 & 1-p_2 & 0 \\ 0 & 0 & 0 & 1-p_3 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

Q is nilpotent ($Q^4 = 0$), so the fundamental matrix is exact and finite:

$$N = (I - Q)^{-1} = I + Q + Q^2 + Q^3$$

N_{ij} = expected number of times the chain occupies transient state j , starting from i . Because each A_n is visited at most once, $N_{1j} \in \{0, 1\}$ along any realized path — it tracks *whether* state j is reached.

1.5 Expected life

$$E[\text{periods to absorption}] = \sum_{n=1}^4 n \cdot \pi_n \cdot a_n$$

where $\pi_n = \prod_{k=1}^{n-1} (1 - p_k)$ is the probability of reaching observation n (with $\pi_1 = 1$), and $a_n = p_n$ for $n < 4$, $a_4 = 1$. Explicitly, in months:

$$E[\text{life}] = t_1 p_1 + t_2 (1 - p_1) p_2 + t_3 (1 - p_1) (1 - p_2) p_3 + t_4 (1 - p_1) (1 - p_2) (1 - p_3)$$

with $t_n \in \{2, 4, 6, 8\}$. This is the expected *redemption date* — the quantity desks call “expected life,” since the product rarely actually survives to the stated 8-month maturity once S has a reasonable chance of rallying.

1.6 Absorption probabilities and expected payoff

$B = NR$ gives the absorption probabilities (i.e, the probability that we end up in a specific outcome s.t $B_{ik} = P(\text{end up absorbed in state } k \text{ start in transient state } i)$; from A_1 these collapse to the simple cascade:

$$P(C_n) = \pi_n p_n, \quad P(M_H) = \pi_4 q_H, \quad P(M_M) = \pi_4 q_M, \quad P(M_L) = \pi_4 q_L$$

1.6.1 Reward structure

This is a Markov chain *with rewards* — coupons attach to transitions, not just terminal states, since a coupon is paid at t_n whenever $S_{t_n} \geq 70\%$, whether or not that triggers a call.

Event	Payoff at t_n
Called at n ($n \leq 3$)	Principal + coupon c
Continues with coupon ($q_n, n \leq 3$)	c only
Continues, no coupon ($r_n, n \leq 3$)	0
M_H	Principal + coupon c
M_M	Principal only
M_L	Principal $\times S_T/S_0$

1.6.2 Undiscounted expected payoff

$$E[\text{Payoff}] = \sum_{n=1}^3 \pi_n \left[p_n (\text{Principal} + c) + q_n c \right] + \pi_4 \left[q_H (\text{Principal} + c) + q_M \text{Principal} + q_L \text{Principal} \cdot E \left[\frac{S_T}{S_0} \mid S_T < 60\% \right] \right]$$

1.6.3 Present-value (discounted) version

Multiply each n -indexed bracket by a discount factor $DF(t_n)$ before summing — this is what you would actually quote as fair value, and it is why expected life matters: it tells you *when*, on average, that PV is realized, which feeds duration/sensitivity analysis even though the PV formula itself does not need the fundamental matrix explicitly.

$$PV = \sum_{n=1}^3 \pi_n DF(t_n) \left[p_n (\text{Principal} + c) + q_n c \right] + \pi_4 DF(t_4) \left[q_H (\text{Principal} + c) + q_M \text{Principal} + q_L \text{Principal} \cdot E \left[\frac{S_T}{S_0} \mid S_T < 60\% \right] \right]$$

1.7 Computing p_n, q_n

Performing this computation requires $p_n = P(S_{t_n} \geq 100\% \mid \text{no autocall at } 1, \dots, n-1)$ — a probability about the *path*, not just the marginal level at t_n . Because GBM is Markov, the clean way to compute these probabilities without approximating with unconditional marginals is to **not** collapse “alive” into a single macro-state, but instead run a discretized price-grid Markov chain: a

lattice/grid over $S_{t_{n-1}}$, propagate through the lognormal transition kernel to t_n , and at each step split the probability mass at the 100% / 70% / 60% boundaries before convolving forward to the next date.

The macro-chain is the right way to *organize* the life/payoff bookkeeping (fundamental matrix, absorption probabilities, expected life); the Monte Carlo sim generates accurate numbers for p_n, q_n, q_H, q_M, q_L given a volatility, rate, and dividend assumption for the underlying.

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